TVET PROGRAMME IMPACT ASSESSMENT REPORT

Using KoboToolbox, CommCare, Google Forms, Microsoft Excel, SQL Server, and Microsoft Power BI

Prepared By

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Data Analytics Bootcamp – Capstone Project

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Executive Summary

Technical and Vocational Education and Training (TVET) programmes play an important role in equipping individuals with practical skills that improve employability, entrepreneurship, and income generation. Measuring the effectiveness of these programmes helps stakeholders understand whether participants are gaining the intended benefits and identify opportunities for improvement.

This project assessed the impact of a TVET programme through a structured survey administered using KoboToolbox, CommCare, and Google Forms. Responses collected from participants were cleaned, validated, and analysed using Microsoft Excel, Microsoft SQL Server, and Microsoft Power BI.

The analysis focused on respondent demographics, training experiences, employment outcomes, programme satisfaction, and participants' recommendations. Descriptive statistics, SQL queries, Pivot Tables, Pivot Charts, and interactive dashboards were used to transform raw survey data into meaningful insights.

The project demonstrates a complete end-to-end data analytics workflow, beginning with digital data collection and ending with interactive reporting. The findings provide practical information that can support programme managers, training institutions, and decision-makers in strengthening future TVET interventions.

1.1 Introduction

Technical and Vocational Education and Training (TVET) programmes are designed to prepare individuals with practical knowledge and technical skills that improve their ability to secure employment or establish sustainable businesses. As governments and development partners continue investing in skills development initiatives, it becomes increasingly important to assess whether these programmes are achieving their intended objectives.

This project applies modern data collection and analytics techniques to evaluate selected aspects of a TVET programme. Digital survey tools were used to collect participant responses, while data analytics tools were applied to clean, analyse, and visualise the results.

Rather than focusing only on survey administration, the project demonstrates the complete lifecycle of a data analytics project—from questionnaire design and data collection to analysis, reporting, and dashboard development.

1.2 Background of the Study

TVET programmes have become an important strategy for addressing unemployment and improving workforce readiness. By providing practical and industry-relevant skills, these programmes aim to increase participants' employability, encourage entrepreneurship, and contribute to economic development.

Despite the growing investment in vocational education, many organisations require evidence showing whether programme participants successfully transition into employment, improve their income, and remain satisfied with the quality of training received.

This assessment was conducted to provide that evidence through structured data collection and analysis. The project combines Monitoring and Evaluation (M&E) principles with modern data analytics techniques to generate findings that can support programme improvement and future planning.

1.3 Problem Statement

Although significant resources are invested in TVET programmes, programme managers often face challenges in measuring their effectiveness after training has been completed. Information on employment outcomes, participant satisfaction, and skill utilisation may not always be readily available or consistently documented.

Without reliable data, it becomes difficult to evaluate programme performance, identify implementation gaps, and make informed decisions for future improvements.

This project addresses that challenge by collecting participant feedback, analysing the responses, and presenting the results through structured reports and interactive dashboards.

1.4 Aim of the Project

The aim of this project is to assess the effectiveness of a Technical and Vocational Education and Training (TVET) programme by collecting, analysing, and visualising participant data using modern digital data collection and business intelligence tools.

1.5 Project Objectives

The specific objectives of this project are to:

- Design a structured digital questionnaire for TVET participants.
- Collect participant responses using KoboToolbox, CommCare, and Google Forms.
- Clean and validate the collected data to ensure accuracy and consistency.
- Perform descriptive statistical analysis using Microsoft Excel.
- Conduct SQL-based analysis using Microsoft SQL Server.
- Develop an interactive Power BI dashboard for data visualisation.
- Generate evidence-based findings and practical recommendations for programme improvement.

1.6 Scope of the Project

This project focuses on assessing selected outcomes of a TVET programme using survey responses obtained from programme participants.

The assessment covers:

- Respondent demographic characteristics
- Training experience
- Skills development
- Employment outcomes
- Income improvement
- Participant satisfaction
- Recommendations for programme improvement

The project demonstrates the practical application of data analytics techniques rather than evaluating the performance of a specific institution or organisation.

Research Methodology

2.1 Research Design

A quantitative survey approach was adopted for this project to assess the impact of the Technical and Vocational Education and Training (TVET) programme. This approach made it possible to collect structured responses from participants and analyse the results using statistical and business intelligence tools. The focus was on gathering measurable information related to participants' backgrounds, training experiences, employment outcomes, and overall satisfaction with the programme.

2.2 Data Collection Process

Data collection was carried out using three digital platforms to demonstrate different approaches to electronic data capture and improve accessibility for respondents.

The survey questionnaire was first developed in **KoboToolbox**, where validation rules, required fields, and skip logic were implemented to improve data quality during data entry.

The same questionnaire was recreated in **CommCare**, allowing the project to demonstrate mobile-based data collection suitable for field environments, including offline functionality.

Finally, **Google Forms** was developed as an additional web-based survey option. This made it easier for respondents to complete the questionnaire using a computer or mobile device, while responses were automatically stored in Google Sheets for easy access.

Using these three platforms provided practical experience with multiple data collection technologies while maintaining consistency across the questionnaire.

2.3 Data Preparation and Cleaning

After data collection, responses were exported into Microsoft Excel for preparation and cleaning.

The cleaning process included reviewing the dataset for missing values, duplicate records, incorrect data types, and unnecessary system-generated variables. Metadata fields that were not required for analysis were removed, while response categories were standardised to improve consistency across the dataset.

The cleaned dataset became the master dataset used throughout the remaining stages of the project.

2.4 Data Quality Assurance

Several quality checks were performed before analysis began.

These included:

- Reviewing the completeness of submitted responses.
- Confirming that there were no duplicate records.
- Checking for missing values in key variables.
- Verifying that numeric and text fields contained valid entries.
- Confirming that skip logic and validation rules worked correctly during data collection.
- Ensuring consistency between exported datasets and the cleaned dataset.

These checks helped improve confidence in the reliability of the analysis.

2.5 Microsoft Excel Analysis

Microsoft Excel was used as the primary tool for exploratory data analysis.

The cleaned dataset was analysed using descriptive statistics, frequency tables, Pivot Tables, and Pivot Charts to summarise the responses. An interactive Excel dashboard was also developed to present key indicators and support quick interpretation of the survey results.

Excel provided the foundation for identifying trends before moving into more advanced analysis.

2.6 SQL Analysis

The cleaned dataset was imported into Microsoft SQL Server for database analysis.

SQL queries were written to retrieve, summarise, and analyse the data using techniques such as filtering, grouping, aggregation, subqueries, Common Table Expressions (CTEs), and window functions. These analyses helped validate the findings obtained from Excel while demonstrating practical SQL skills.

2.7 Power BI Dashboard Development

The final stage of the analytical process involved building an interactive dashboard in Microsoft Power BI.

The cleaned dataset was imported into Power BI, where additional transformations were completed using Power Query. DAX measures were created to calculate key performance indicators, while charts, slicers, and cards were used to build an interactive dashboard that allows users to explore the survey results from different perspectives.

The dashboard provides a clear visual summary of the programme's performance and key outcomes.

2.8 Ethical Considerations

Participation in the survey was voluntary. Respondents were informed about the purpose of the assessment before providing their responses. Personal information collected during the survey was used only for the purpose of this project, and the analysis focused on aggregated results rather than individual responses.

2.9 Summary

The methodology adopted in this project combined digital data collection, structured data preparation, statistical analysis, SQL-based querying, and business intelligence reporting. Using multiple tools throughout the workflow provided an opportunity to demonstrate practical skills across different stages of the data analytics process while producing reliable and meaningful insights.

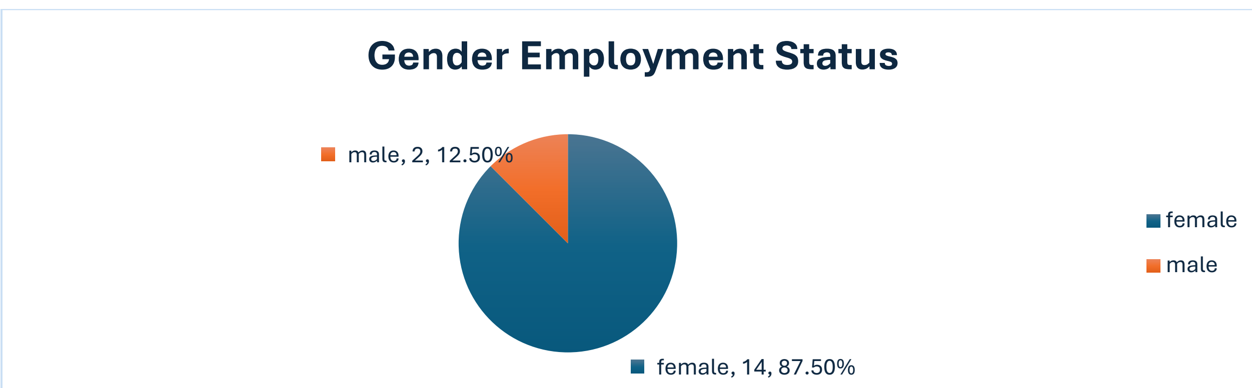
3.0 Results and Findings

This chapter presents the results of the analysis carried out on the survey data collected for the TVET Programme Impact Assessment. The findings are based on **54 valid responses** obtained from the cleaned dataset. Microsoft Excel, SQL Server, and Microsoft Power BI were used to analyse and visualise the data, while Pivot Tables and interactive dashboards were used to identify key patterns and trends.

The findings are presented under different themes, including respondent demographics, employment outcomes, training experience, programme satisfaction, and participants' recommendations.

3.1 Respondent Demographics

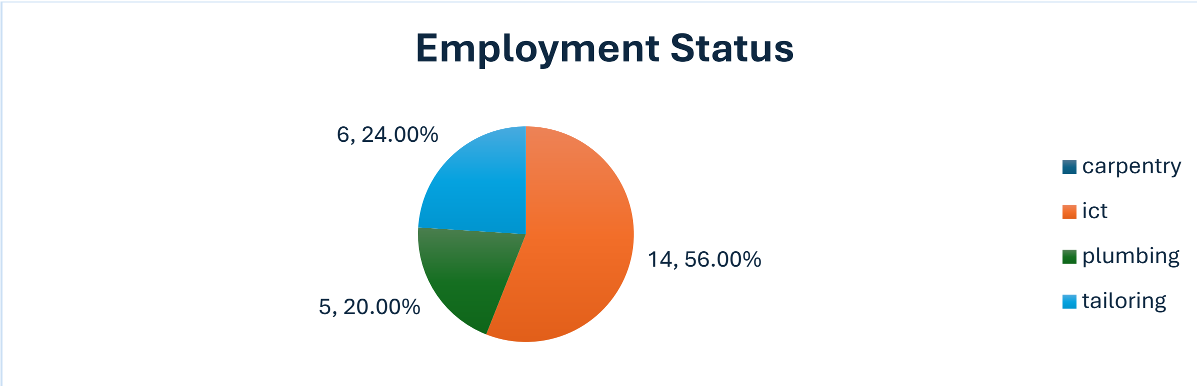
The final dataset contained **54 valid responses**. Of these, **30 respondents (55.6%) were male**, while **24 respondents (44.4%) were female**. Although male respondents formed a slightly larger proportion of the sample, the distribution shows that both genders were reasonably represented in the assessment.



3.2 Employment Status of Respondents

Employment outcomes were assessed to understand the extent to which participants were engaged in economic activities after completing the programme. The analysis shows that **38 respondents (70.4%) were self-employed**, while **16 respondents (29.6%) were employed**.

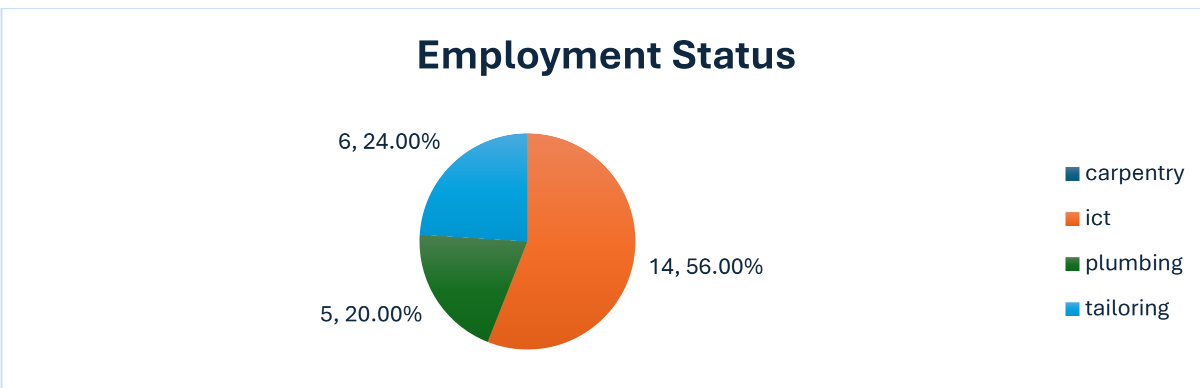
The higher proportion of self-employed respondents suggests that many participants are applying the vocational skills acquired during the programme to create their own sources of income rather than relying solely on formal employment opportunities.



3.3 Education Level and Employment Status

The relationship between respondents' educational qualifications and their employment status was examined using a Pivot Table.

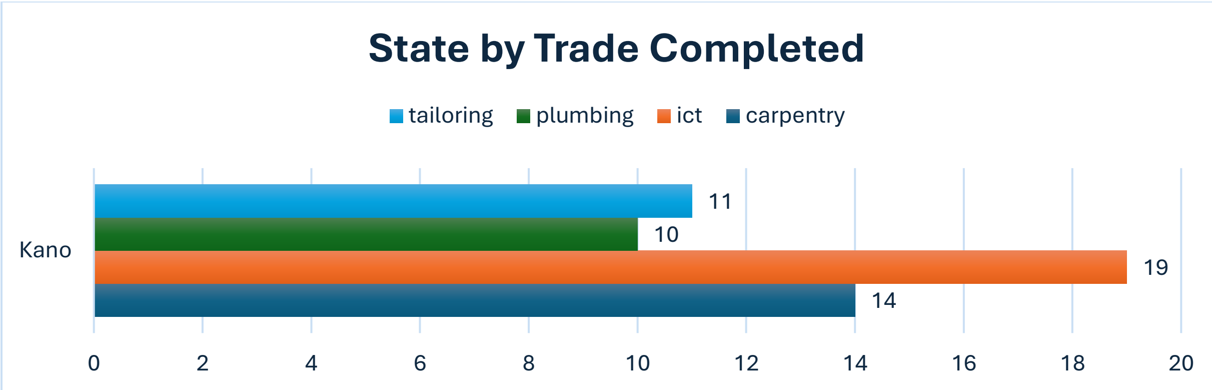
The results indicate that respondents from different educational backgrounds were represented among both employed and self-employed groups. This suggests that the programme provides opportunities for participants regardless of their previous educational attainment, although employment outcomes may vary across education levels.



3.4 Trade Completed and Employment Status

The analysis also explored employment outcomes across different vocational trades completed by respondents.

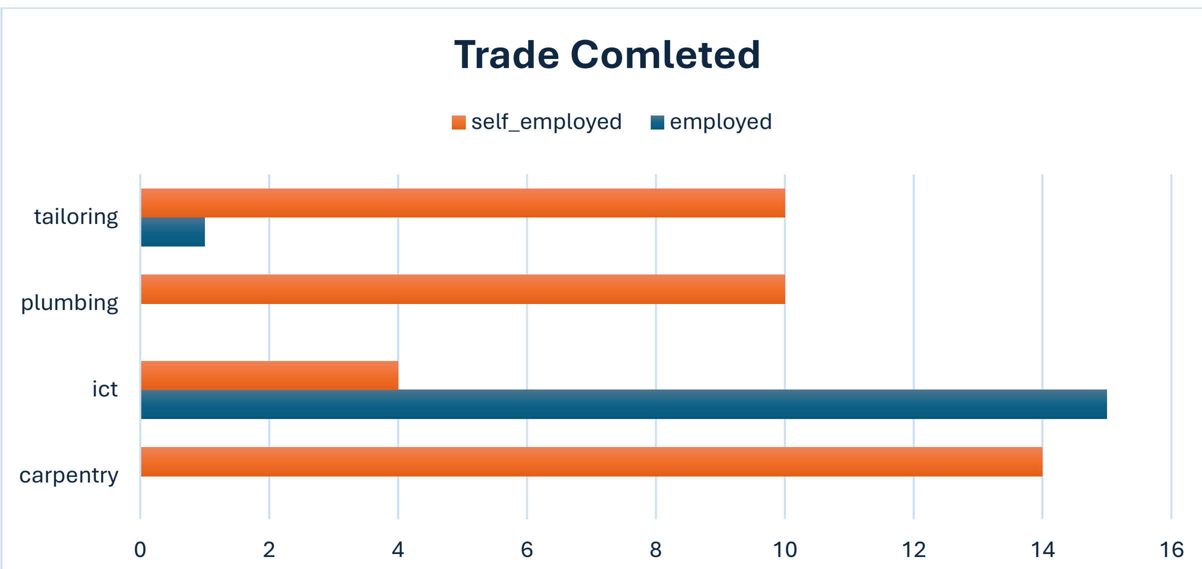
Participants who completed different training programmes were represented in both employment categories. The findings indicate that vocational training contributes to employability across multiple skill areas, although some trades appear to have higher levels of self-employment than others.



3.5 Trade Completed and Income Improvement

Respondents were asked whether completing the training had improved their income.

The analysis indicates that participants from different vocational trades generally reported positive changes in their income after completing the programme. This finding suggests that the practical skills acquired during training have contributed to improved livelihood opportunities.



3.6 Overall Satisfaction with the Programme

Participant satisfaction was assessed to understand how respondents perceived the quality of the training received.

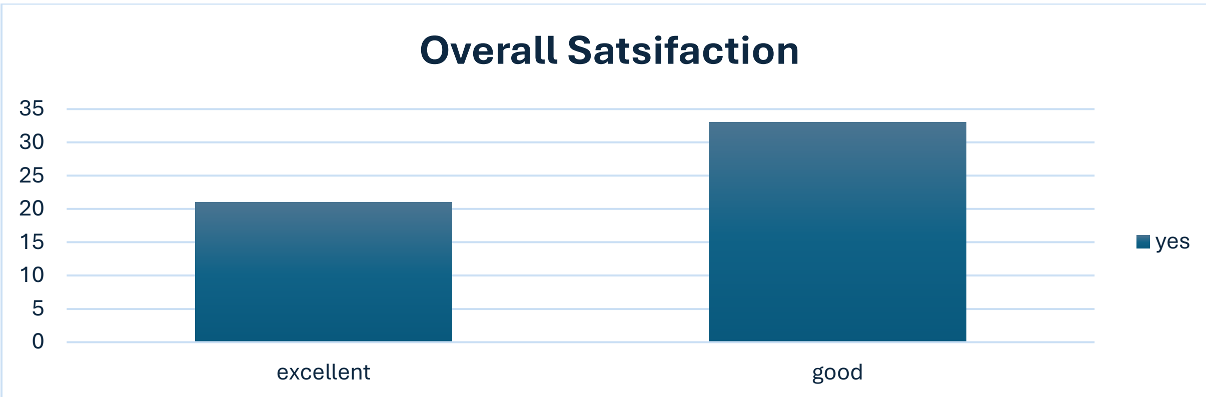
The responses show a consistently positive assessment of the programme. Respondents rated their overall experience as either **Good** or **Excellent**, indicating a high level of satisfaction with the quality of training, instructors, facilities, and learning experience.

Insert Figure 3.3: Overall Satisfaction

3.7 Recommendation of the Programme

Respondents were asked whether they would recommend the programme to others.

The findings show a very positive response, with participants expressing a strong willingness to recommend the programme. This reflects confidence in the value of the training and suggests that participants believe others could also benefit from similar opportunities.



3.8 Interest in Future Participation

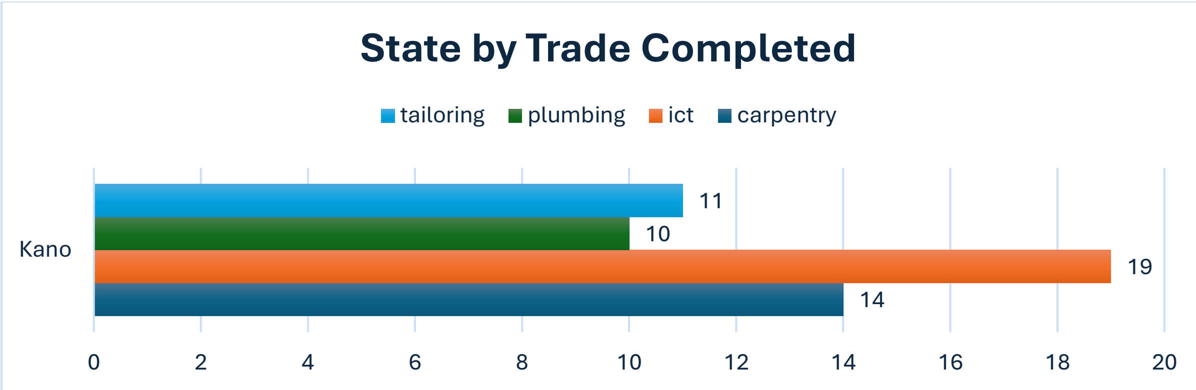
Participants were also asked whether they would be willing to participate in similar programmes in the future.

The responses indicate a high level of continued interest. This suggests that participants found the training useful and would welcome additional opportunities for skills development and professional growth.

3.9 Distribution of Trades Across States

The analysis examined the distribution of vocational trades across the participating state.

The results show that respondents completed a variety of vocational trades, demonstrating the diversity of skills represented within the programme. This provides useful information for understanding programme coverage and planning future training interventions.



3.10 Summary of Findings

The analysis highlights several important observations from the survey. The respondent population included both male and female participants, with a slightly higher proportion of males. Most respondents reported being self-employed after completing the programme, suggesting that the training has supported entrepreneurship and independent income generation.

Participants expressed positive views about the programme, with high levels of satisfaction and a strong willingness to recommend it to others. Overall, the findings suggest that the TVET programme has made a positive contribution to participants' skills development and employment outcomes.

4.1 Discussion of Findings

The findings from this assessment provide valuable insights into the outcomes of the TVET programme and its contribution to participants' skills development and employment.

One of the key observations is the balanced participation of both male and female respondents. Although male respondents made up a slightly larger share of the sample, the presence of both genders suggests that the programme was accessible to a diverse group of participants. This is important because inclusive participation strengthens the relevance and reach of vocational training initiatives.

The employment analysis shows that a considerable proportion of respondents were self-employed after completing the programme. This suggests that many participants were able to apply the practical skills they acquired to create income-generating opportunities. While formal employment remains an important outcome, the results highlight the role of vocational education in supporting entrepreneurship and self-reliance.

The relationship between education level and employment status indicates that participants from different educational backgrounds benefited from the programme. This reinforces the idea that vocational training can complement existing education and provide practical skills that improve employability regardless of previous academic qualifications.

Another important finding relates to income improvement. Responses across different vocational trades suggest that many participants experienced positive economic outcomes after completing their training. Although the level of improvement may differ between trades, the overall pattern indicates that the programme has contributed to better livelihood opportunities.

Participants also expressed positive opinions about the quality of the training. High satisfaction ratings, together with the willingness to recommend the programme to others and participate in similar initiatives in the future, reflect confidence in the value of the training experience. Positive feedback of this nature is often associated with effective programme delivery, relevant course content, and supportive learning environments.

Overall, the findings suggest that the TVET programme has had a positive influence on participants by improving practical skills, supporting employment opportunities, and encouraging entrepreneurship. These outcomes demonstrate the importance of continued investment in vocational education as a strategy for economic empowerment and workforce development.

4.2 Implications of the Findings

The results of this assessment have practical implications for organisations involved in vocational skills development. First, the strong level of participant satisfaction indicates that maintaining high training standards should remain a priority. Continuous investment in qualified instructors, modern training equipment, and practical learning opportunities can help sustain these positive outcomes.

Second, the high proportion of self-employed participants highlights the need for complementary support services such as business mentoring, access to finance, and entrepreneurship development. Providing these additional resources could help graduates expand their businesses and improve long-term income generation.

Finally, regular monitoring and evaluation should continue to be integrated into TVET programmes. Collecting and analysing participant feedback enables programme managers to identify strengths, address implementation challenges, and make informed decisions that improve future training cycles.

4.3 Chapter Summary

This chapter discussed the main findings of the assessment and explored their practical significance. The results indicate that the TVET programme has contributed positively to skills development, employment, and participant satisfaction. The findings also highlight opportunities for strengthening future programme implementation through continued investment in training quality, entrepreneurship support, and evidence-based decision-making.

5.1 Conclusion

This project assessed the impact of the Technical and Vocational Education and Training (TVET) programme using data collected from programme participants through KoboToolbox, CommCare, and Google Forms. The collected responses were cleaned, analysed, and visualised using Microsoft Excel, Microsoft SQL Server, and Microsoft Power BI.

The findings indicate that the programme has made a positive contribution to participants' skills development and employment outcomes. Many respondents reported being engaged in self-employment after completing the training, while others secured paid employment. Participants also expressed positive opinions about the quality of the training and showed a strong willingness to recommend the programme to others.

Beyond the survey findings, this project demonstrates a complete data analytics workflow, covering digital data collection, data cleaning, statistical analysis, database querying, and dashboard development. It highlights how data can be transformed into meaningful information that supports planning, monitoring, and decision-making.

Overall, the project achieved its objectives and provides a practical example of how data analytics can be applied to evaluate development programmes and support continuous improvement.

5.2 Recommendations

Based on the findings of this assessment, the following recommendations are proposed:

- Continue investing in practical and hands-on training to strengthen participants' technical skills.
- Expand entrepreneurship support services, including business mentoring and access to startup resources, to help graduates sustain self-employment.
- Maintain regular monitoring and evaluation activities to track programme performance and measure long-term outcomes.
- Review training curricula periodically to ensure that the skills being taught remain relevant to current labour market needs.
- Strengthen partnerships with employers, industries, and vocational institutions to increase employment opportunities for graduates.
- Encourage periodic feedback from participants to support continuous improvement of programme delivery.

5.3 Challenges Encountered

A number of challenges were encountered during the implementation of this project:

- Limited time available for collecting survey responses before the project deadline.
- Difficulty obtaining a larger number of respondents within the available timeframe.
- Managing data collected from multiple survey platforms while maintaining consistency across datasets.
- Ensuring that all survey forms contained the same structure, validation rules, and skip logic.

- Cleaning and preparing the data before analysis required careful review to maintain data quality.

Despite these challenges, the project was completed successfully through careful planning, systematic data management, and the use of appropriate analytical tools.

5.4 Lessons Learned

This project provided valuable practical experience across the complete data analytics lifecycle. It strengthened skills in digital data collection, data preparation, exploratory analysis, SQL querying, dashboard development, and technical reporting.

The project also highlighted the importance of data quality, documentation, and effective communication in producing reliable analytical results that can support evidence-based decision-making.

References

The following software and resources were used during the project:

- KoboToolbox. (2026). *Data Collection Platform*.
- CommCare by Dimagi. (2026). *Mobile Data Collection Platform*.
- Microsoft Excel. Microsoft Corporation.
- Microsoft SQL Server. Microsoft Corporation.
- SQL Server Management Studio (SSMS). Microsoft Corporation.
- Microsoft Power BI. Microsoft Corporation.
- Google Forms. Google LLC.

Appendix

The appendix contains supporting materials developed during the project, including:

- Survey questionnaire
- KoboToolbox form design
- CommCare application structure
- Google Form
- Data Quality Report
- Pivot Tables

- Pivot Charts
- SQL queries
- Power BI dashboard screenshots
- Additional project documentation